



Starfish (an echinoderm)

**GUYOT** Flat-topped seamount that once rose above the ocean's surface as a volcanic island and whose surface was eroded by wind and waves. (see also SEAMOUNT)

**HURRICANE** Tropical storm with winds of over 74 mph (119 km/h), forming over the Atlantic Ocean. Tropical storms are usually called typhoons in the Pacific Ocean and cyclones in the Indian Ocean.

**ICEBERG** Floating mass of ice broken off an ice sheet or glacier, carried along by ocean currents.

**INVERTEBRATE** Animal without a backbone.

**KRILL** Shrimplike crustacean that lives in polar waters of the Arctic and Antarctic in great numbers, forming much of the food supply of baleen whales.

**MAGMA** Molten rock that lies beneath the Earth's crust.

**MARINE BIOLOGY** The study of ocean life.

**MIDOCEAN RIDGE** Long, undersea mountain range forming where two tectonic plates are pulling apart, with magma rising from beneath the Earth's surface and hardening into rock.

**MOLLUSK** An invertebrate with a soft body usually enclosed by a shell. Mollusks include bivalves (such as clams), gastropods (such as sea slugs), and cephalopods (such as squid and octopuses). (see also BIVALVE, CEPHALOPOD)

**OCEANOGRAPHY** The scientific study of the oceans.

**PHYTOPLANKTON** Microscopic single-celled algae that drift in the ocean's sunlit zone. (see also SUNLIT ZONE)

**PLANKTON** Tiny plant and animal organisms that drift in the sea's surface waters, providing the basis of most marine food chains. (see also FOOD CHAIN, ZOOPLANKTON)

**PLATE TECTONICS** The study of the movement of the Earth's lithospheric plates that carry the oceanic and continental crust.

**POLYP** A sea anemone or coral with a mouth surrounded by tentacles. A hard coral polyp makes a limestone cup, or skeleton, to protect its body. Thousands of polyps live together in colonies, forming a coral reef.

**ROV** Short for Remotely Operated Vehicle—a small vessel operated from (and tethered to) a submersible or ship.

**SALINITY** The amount of dissolved salt in seawater. Salinity is measured as parts of salt per 1,000 parts of seawater, the average salinity of the oceans being 35 parts of salt per 1,000 parts of seawater.

**SCUBA** Stands for Self-Contained Underwater Breathing Apparatus—SCUBA divers carry their own air supply in tanks on their backs.

**SEA** Another word for ocean, or a particular part of an ocean—for example, the Black Sea and the Mediterranean Sea are connected to the Atlantic Ocean.

**SEAMOUNT** Underwater volcano that rises 3,300 ft (1,000 m) or more above the surrounding plain.

**SEDIMENT** Mud, sand, and silt, containing millions of tiny plants and animals, washed off the land by rivers. Sediment settles on the ocean floor.

**SONAR** Short for Sound Navigation And Ranging—a system that can locate the position of an object by emitting sounds then timing the echoes that bounce back.

**SUBMERSIBLE** Manned or remotely operated underwater research submarine designed to withstand water pressure in deep water. (see also WATER PRESSURE)

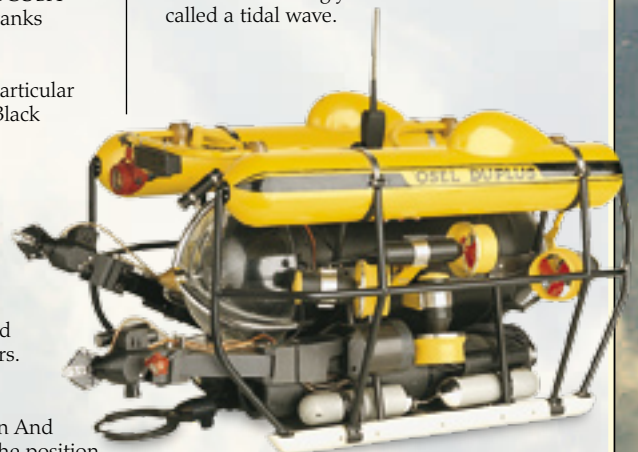
**SUNLIT ZONE** Surface layer of the ocean penetrated by sunlight, to around 650 ft (200 m) deep. Most marine life lives in this zone. Also called the epipelagic zone.

**SYMBIOSIS** Close interaction between two different species where either, both, or neither benefit from the relationship.

**TIDE** The regular rise and fall of the sea caused by the gravitational pull of the Sun and the Moon on the Earth.

**TRENCH** A steep-sided trough or valley in the ocean floor.

**TSUNAMI** Sea wave usually caused by an underwater volcanic eruption or earthquake. It can cause great damage if it reaches the coast as it may gain considerable height in shallow water. Sometimes wrongly called a tidal wave.



Submersible

**TWILIGHT ZONE** Area of the ocean from around 650–3,300 ft (200–1,000 m) deep, bordered by the sunlit zone above and darkness below. Also called the mesopelagic zone.

**TYPHOON** Name given to a tropical storm in the western Pacific Ocean. (see also HURRICANE)

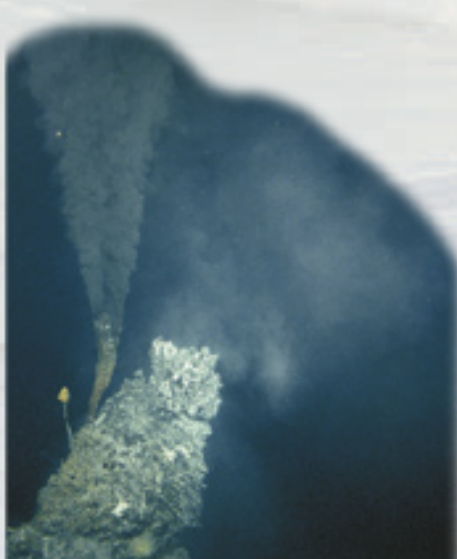
**UPWELLING** Rising of nutrient-rich water from deeper parts of the ocean to the surface.

**WATER PRESSURE** Force exerted by water because of its weight and density; water pressure increases by one atmosphere for each 33 ft (10 m) depth.

**WAVE HEIGHT** The distance between the crest (top of a wave) and its trough (lowest part of a wave).

**WAVELENGTH** The vertical distance between two successive wave crests (the tops of the waves).

**ZOOPLANKTON** Tiny animals that float in the water, such as copepods and tiny crustaceans, forming part of plankton. (see also PHYTOPLANKTON, PLANKTON)



Black smoker